Topic 1 End of Topic Assessment: Answer Key

Original (standard) form

ANSWER KEY

Multiple-choice letters match the original test (A–D on odd questions, F–J on even). Suggested points: 1 per multiple-choice item; 2 per written item (1 for the value, 1 for the units or the explanation). Total: **21 points**.

Q	Answer	Work / explanation	Pts
1	**C.** y = 1750x + 40,500	x = 0, 1, 2, 3 (years since 2000); y = salary. LinReg(ax+b) gives a = 1750, b = 40,500.	1
2	**J.** When you don't brush your teeth, you get cavities.	Not brushing causes cavities. F: being at the beach doesn't make you wet (you might stay dry). G: correlation only. H: the umbrella doesn't make it rain (reversed).	1
3	2.75 dollars per cookie	(1.25 − (−1.50)) ÷ (5 − 4) = 2.75 ÷ 1 = 2.75. Every pair of rows rises by 2.75, so the rate is constant: Nia's profit goes up $2.75 for each cookie sold.	2
4	**C.** Weak positive correlation	LinReg with month in L1 and temperature in L2 gives r ≈ 0.04. It is positive but close to 0, so weak (the temperatures rise and then fall, which isn't linear).	1
5	y = −77x + 950	x = 0 in March, so b = 950. Each month: 873 − 950 = −77 (constant first differences), so m = −77.	2
6	**H.** $131.46	LinReg: y ≈ 14.357x + 2.25. At x = 9: 14.357(9) + 2.25 ≈ 131.46.	1
7	−5	3y = −15x + 45 → y = −5x + 15, so m = −5. (Or m = −A/B = −15/3.)	1
8	y = (1/3)x − 5/3	Point-slope: y − (−2) = (1/3)(x − (−1)) → y + 2 = (1/3)x + 1/3 → y = (1/3)x + 1/3 − 6/3 = (1/3)x − 5/3.	2

9	−60 liters per minute	(240 − 360) ÷ (2 − 0) = −60. Check: (150 − 240) ÷ 1.5 = −60, (60 − 150) ÷ 1.5 = −60. **Meaning:** the water level goes down 60 liters every minute.	2
10	(40, 0)	The line meets the x-axis at 40 minutes. **Meaning:** after 40 minutes Gracie is 0 miles from home, so she has arrived home.	2
11	(0, 2 2/3), about (0, 2.67)	Each box on the y-axis is 1/6 mile; the line starts 4 boxes above 2. **Meaning:** when Gracie leaves school (0 minutes) she is 2 2/3 miles from home, so school is 2 2/3 miles from home.	2
12	−1/15 mile per minute	Points (10, 2) and (25, 1): (1 − 2) ÷ (25 − 10) = −1/15. Or (0 − 8/3) ÷ (40 − 0) = −1/15. **Meaning:** Gracie gets 1/15 mile closer to home each minute (4 miles per hour).	2
13	**B.** −94.94	f(−4.2) = 23.2(−4.2) + 2.5 = −97.44 + 2.5 = −94.94.	1
14	**G.** {0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10}	Domain = x-values (batches). Miguel bakes only whole batches and runs out of eggs after 10, so the domain is discrete. F includes fractions of a batch; H and J describe y (eggs), not x.	1

Quick key

1 C · 2 J · 3 $2.75/cookie · 4 C · 5 y = −77x + 950 · 6 H · 7 −5 · 8 y = (1/3)x − 5/3 · 9 −60 L/min · 10 (40, 0) · 11 (0, 2 2/3) · 12 −1/15 mi/min · 13 B · 14 G

Form M (504/IEP) uses the same answers; only the multiple-choice letters change: 1 B · 2 C · 4 B · 6 B · 13 B · 14 B. See the Form M answer key for the filled-in student pages.